

Quadratic Arithmetic Programs (QAP)

Pushpendra Pal

June 2025

Keywords: Zero knowledge proof, SNARK, Groth, KZG, PCS, polynomial commitment scheme, rank-1 constraint system, R1CS, QAP

1 R1CS to QAP Example: 2 Constraints, 4 Variables

Circuit Example : $P(x) = x*y + x$

Let's prove we know values that satisfy:

$x * y = z$ (constraint 1)
 $z + x = w$ (constraint 2)

With witness: $x = 3, y = 2, z = 6, w = 9$

1.1 Step 1: R1CS Representation

Variables ordering: $[1, x, y, z, w] = [1, 3, 2, 6, 9]$ Matrix A (left operands):

$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$ #x *
#(x+z)

Matrix B (left operands):

$B = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$ # .. *y
.. *1

Matrix C (left operands):

$C = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ # = z
= w

1.2 Step2: Verify R1CS

Let $s = [1, 3, 2, 6, 9]$ Constraint 1: $(A_1 \cdot s) \times (B_1 \cdot s) = (C_1 \cdot s)$

- $A_1 \cdot s = 0x1 + 1x3 + 0x2 + 0x6 + 0x9 = 3$
- $B_1 \cdot s = 0x1 + 0x3 + 1x2 + 0x6 + 0x9 = 2$
- $C_1 \cdot s = 0x1 + 0x3 + 0x2 + 1x6 + 0x9 = 6$
- Check: $3 \times 2 = 6$

Constraint 2: $(A_2 \cdot s) \times (B_2 \cdot s) = (C_2 \cdot s)$

- $A_2 \cdot s = 0x1 + 1x3 + 0x2 + 1x6 + 0x9 = 9$
- $B_2 \cdot s = 1x1 + 0x3 + 0x2 + 0x6 + 0x9 = 1$
- $C_2 \cdot s = 0x1 + 0x3 + 0x2 + 0x6 + 1x9 = 9$
- Check: $9 \times 1 = 9$

1.3 Step3: Choose Evaluation Points

Instead of using 0, 1, let's use: 2, 5 (random field elements)

1.4 Step4: Create QAP Polynomials

For each column, we interpolate a polynomial through the evaluation points. A polynomials:

- $A_0(x)$ (column 0): Points (2,0), (5,0) $\rightarrow A_0(x) = 0$
- $A_1(x)$ (column 1): Points (2,1), (5,1) $\rightarrow A_1(x) = 1$
- $A_2(x)$ (column 2): Points (2,0), (5,0) $\rightarrow A_2(x) = 0$
- $A_3(x)$ (column 3): Points (2,0), (5,1) $\rightarrow A_3(x) = (x-2)/3$
- $A_4(x)$ (column 4): Points (2,0), (5,0) $\rightarrow A_4(x) = 0$

B polynomials:

- $B_0(x)$ (column 0): Points (2,0), (5,1) $\rightarrow B_0(x) = (x-2)/3$
- $B_1(x)$ (column 1): Points (2,0), (5,0) $\rightarrow B_1(x) = 0$
- $B_2(x)$ (column 2): Points (2,1), (5,0) $\rightarrow B_2(x) = (5-x)/3$
- $B_3(x)$ (column 3): Points (2,0), (5,0) $\rightarrow B_3(x) = 0$
- $B_4(x)$ (column 4): Points (2,0), (5,0) $\rightarrow B_4(x) = 0$

C polynomials:

- $C_0(x)$ (column 0): Points (2,0), (5,0) $\rightarrow C_0(x) = 0$
- $C_1(x)$ (column 1): Points (2,0), (5,0) $\rightarrow C_1(x) = 0$
- $C_2(x)$ (column 2): Points (2,0), (5,0) $\rightarrow C_2(x) = 0$
- $C_3(x)$ (column 3): Points (2,1), (5,0) $\rightarrow C_3(x) = (5-x)/3$
- $C_4(x)$ (column 4): Points (2,0), (5,1) $\rightarrow C_4(x) = (x-2)/3$

1.5 Step5: QAP Condition

The QAP requires that there exists a polynomial $H(x)$ such that:

$$A(x) \times B(x) - C(x) = H(x) \times Z(x)$$

Where:

$$\begin{aligned} A(x) = \sum_i s_i \times A_i(x) &= 3x A_1(x) + 6x A_3(x) \\ &= 3x1 + 6x(x-2)/3 \\ &= 3 + 2(x-2) \\ &= 2x - 1 \end{aligned}$$

$$\begin{aligned} B(x) = \sum_i s_i \times B_i(x) &= 1x B_0(x) + 2x B_2(x) \\ &= (x-2)/3 + 2x(5-x)/3 \\ &= (x-2+10-2x)/3 \\ &= (8-x)/3 \end{aligned}$$

$$\begin{aligned} C(x) = \sum_i s_i \times C_i(x) &= 6x C_3(x) + 9x C_4(x) \\ &= 6x(5-x)/3 + 9x(x-2)/3 \\ &= (30-6x+9x-18)/3 \\ &= (12+3x)/3 \\ &= x+4 \end{aligned}$$

$$\begin{aligned} Z(x) &= (x-2)(x-5) \\ &= x^2 - 7x + 10 \end{aligned}$$

1.6 Step6: Verify QAP

$$A(x) \times B(x) - C(x):$$

$$\begin{aligned} A(x) \times B(x) &= (2x-1) \times (8-x)/3 \\ &= (2x-1)(8-x)/3 = (16x-2x^2-8+x)/3 \\ &= (17x-2x^2-8)/3 \end{aligned}$$

$$\begin{aligned} A(x) \times B(x) - C(x) &= (17x-2x^2-8)/3 - (x+4) \\ &= (17x-2x^2-8-3x-12)/3 \\ &= (14x-2x^2-20)/3 \\ &= -2(x^2-7x+10)/3 \end{aligned}$$

$$\text{Therefore: } H(x) = -2/3$$

Since $A(x) \times B(x) - C(x) = H(x) \times Z(x)$, the QAP is satisfied!

Key Insights

- Evaluation points 2, 5 were chosen arbitrarily (not 0, 1)
- Polynomial interpolation creates unique degree-1 polynomials through 2 points
- The QAP condition is satisfied, proving the original constraints hold
- Security: The prover knows $s = [1, 3, 2, 6, 9]$ that satisfies both the R1CS and QAP.

2 Questions

- Why do we use random points instead of just indices of the rows for Lagrange interpolation evaluation points? **ANS.** Evaluation Points Choice

The evaluation points are usually:

- Random field elements or
- Roots of unity (when using FFT-friendly constructions)
- Not simply the row indices 0, 1, 2, 3, ...

Why? Because:

- Security: Random points provide better security properties in the cryptographic protocol.
- Efficiency: Roots of unity enable fast polynomial operations using FFT
- Field arithmetic: We're working in a finite field, so we want points that are well-distributed in that field

The choice of evaluation points affects:

- Interpolation: Lagrange interpolation works with any distinct points
 - Protocol efficiency: Some choices enable faster computations
 - Security: Random points prevent certain attacks
-
- What is the rank in rank 1 constraint system? **ANS.** Rank-1 indicates that there is only one multiplication per constraint.